

Empirical Lawhood

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Abstract

How can reliable scientific capabilities accumulate when the conditions needed to acquire and use knowledge must themselves be discovered or constructed? Motivated by dependable fusion propulsion for a journey to Neptune, Empirical Lawhood investigates how knowledge can help establish the means of further discovery and action. Kant’s inquiry into warrant, Wheeler’s observer-participancy and Gell-Mann and Hartle’s physical account of regularity ground an investigation in which observers, instruments and the conditions of inquiry belong to the world being investigated. The framework is developed and applied through human-directed scientific discovery and prospective tests of particular constructions. Analytical and numerical resistor–capacitor studies expose consequential distinctions omitted by reduced observations. Matrix investigations separate geometric organisation from dynamical commitment, an approximate algebra from its scalar detector, and accurate intervention-induced changes from uncertain absolute futures. Two prospective constructions then establish different connections. A forecast supplies the input to fixed response coefficients, supporting both requested assays in 61 of 64 fresh episodes after separate calibration of the composition. A selected preparation makes an entirely unchanged response package usable on 7,965 of 8,192 request pairs across 32 fresh local episodes, versus none under unchanged preparation; a conservative numerical check retains 6,720 pairs. The dominant gain is qualified applicability, not newly attainable response. Further simulations and physical records locate limits of information, uncertainty, delivery and transfer. These findings are scientific outputs of the inquiry, not only tests of a general framework. They support a programme for cumulative capability: a civilisational response-law engine whose missing prerequisites become explicit scientific targets. A further aim is to broaden scientific agency—people’s capacity to choose and redirect inquiry—without making autonomous agenda-setting a condition of using automated capabilities.

1 The conditions of dependable capability

This inquiry began with a practical ambition: what would make fusion propulsion dependable enough for a journey to Neptune? The aspiration was a “Toyota in space”—reliability rather than an isolated spectacular performance. A dependable spacecraft would require more than separately impressive components. Materials, preparation, measurement and control would have to supply one another’s requirements across changing operating conditions. The attempt therefore became a more general scientific construction: acquire response knowledge, determine when it supports an intended use, and investigate the capabilities needed to make that use possible. No propulsion design is offered here; that demand set the scale of the question.

Empirical Lawhood investigates how response knowledge is physically realised, empirically warranted and made reusable—including cases in which another operation can supply a missing condition of its use. A useful model may require an unavailable observation, a discarded distinction or a preparation not yet achieved. The inquiry asks which requirements matter and whether further qualified operations can supply them. This can guide open-ended discovery as well as action towards a specified target.

The empirical result is already concrete. A forecast was used to choose actions through fixed response coefficients before the relevant preparation occurred. In a separate experiment, preparation changed the system so that an entirely unchanged response package became usable. The latter produced a large gain in *qualified applicability*, while post hoc tests showed that suitable physical responses were already widely

available without it. Studies of reduced observations, matrix organisation and transient response explain why these distinctions matter (Table 1).

Study and question	What was tested	Result and meaning
Information: what must a description preserve?	Hidden circuit states with equal analytical inputs or tolerance-matched numerical histories.	An exact error bound and 1,524 response-adverse numerical pairs, including 899 decision-adverse pairs. Numerical witnesses are nested in 36 seed blocks and are not a prevalence estimate.
Structure and prediction: what follows from an apparent success?	Matrix geometry, structural diagnostics, directional prediction and preparation-induced changes.	Structure, dynamical commitment and predictive adequacy separate. Accurate changes do not locate an uncertain absolute future.
Forecast-mediated reuse: can another predictor supply a required input?	A predicted prepared-state description feeds fixed response coefficients; choices precede preparation.	Both tasks succeed in 61/64 fresh episodes. The composition is separately calibrated; task superiority over the cached comparator is not established.
Preparation-mediated applicability: can action supply a required condition?	Physical preparation changes while the entire response package stays fixed.	7,965/8,192 qualified successful pairs, versus zero without that preparation; 6,720 under a conservative numerical check. The principal effect is applicability, not new task capacity.

Table 1: The empirical argument in one view. These are distinct studies, not successive stages of one demonstrated controller. Numerical pairs, task requests and simulation branches remain nested within their study-specific populations. The two prospective constructions supply different prerequisites and have not been joined into one qualified preparation-selection-and-use chain. Detailed results and sources appear in §§4–6. [18–22]

A second motivation concerns scientific agency: the ability to originate questions, choose investigative paths and revise them in light of evidence. The intended engine should enlarge that capacity, not merely deliver discoveries on behalf of people who cannot influence their direction. Automation can supply much of the work without determining the purposes of inquiry. This is a design aim of the programme, distinct from the empirical results above.

1.1 A hammer, a repair, a bench

Consider someone making a bench. The hammer’s adequacy belongs to an activity: this nail, this timber, this grip and this intended joint. Heidegger’s account of equipment makes that practical involvement explicit, including the way breakdown brings otherwise unobtrusive dependencies into view. The following extension concerns the construction of scientific capability. [16, §§15–16]

The nail bends. Striking harder may not help: perhaps the nail is unsuitable, the timber needs a pilot hole, or the unsupported workpiece moves with the blow. A background condition becomes an explicit problem. Supporting the timber changes what the same hammering can accomplish. Making the support requires further materials and capabilities. The completed bench can then provide a stable surface for later work. An activity has produced not only its immediate object but a condition of another activity.

Calling the joint secure does not secure it. Conversely, securing it changes the world rather than merely adding a proposition about an unchanged object. The distinction carries into the experiments: a good description, evidence warranting its use and an operation producing a required state are different achievements. The example makes the dependency intelligible; whether a particular scientific requirement can actually be supplied is an empirical question.

Statistical mechanics and open, driven-dissipative systems provided the starting perspective. A tractable matrix medium allowed preparation, hidden states, observation choices and alternative futures to be investigated together. Physical records tested what available measurements and interventions supported.

The metatheory and investigations developed together: the framework helped formulate questions, while their outcomes changed the framework and the next questions pursued. The resulting work is both an application of EL to scientific discovery and an investigation of its proposed connections.

2 Warrant, participation and physical realisation

2.1 Kant: what entitles a description to function as knowledge?

Kant's critical project concerns the possibility, authority and limits of cognition, including how representations can constitute experience of objects. In the transcendental deduction, the justification of a concept's application is distinguished from the history of its acquisition. The conditions of possible experience delimit legitimate employment of the categories; particular empirical laws still require experience. An account of how a representation arose is therefore not, by itself, an account of its objective validity. [1, B19, A84/B116, A158/B197, B165]

The inheritance taken up here is the demand to make entitlement explicit. For a particular scientific use, what connects this description to its object, and what does the evidence permit it to support? Predicting a mean, identifying a response relation, qualifying uncertainty and justifying a choice are not interchangeable accomplishments. The empirical inquiry consequently tests them separately. Their mathematical and experimental distinctions do not require Kant as a premise; his critical question determines why entitlement, rather than successful association alone, is a central object of the inquiry.

The move is to contingent conditions of particular uses, not to transcendental conditions renamed as experimental variables. A sensor or calibration set is not a category of understanding, and a successful predictor is not a transcendental deduction. Explaining the physical production of an instrument or observer likewise does not, on its own, justify claims made through it. The experiments must establish the relevant connections to their objects.

2.2 Wheeler: how does asking a physical question participate in its answer?

Wheeler's *observer-participancy* gives the choice and physical implementation of a question a substantive role. In *Law without Law*, his discussion of measurement connects what can be asserted about a phenomenon to the apparatus through which an answer is registered. The investigator does not merely receive a description from outside the world: arranging the experiment is part of the phenomenon's specification. [25, pp. 182–187]

It from bit advances a stronger conjecture: physical entities, and ultimately physics itself, are to be understood through apparatus-elicited answers to physical questions. The self-excited circuit depicts a universe giving rise to observers whose participation, in Wheeler's proposal, helps constitute its physical reality. This is not simply a metaphor for feedback control. It is a conjecture about the origins of physical reality and law. [25, pp. 209–210; 26]

For EL, the productive inheritance is to make the question, apparatus and registered consequence explicit parts of the scientific object, and then investigate the conditions that allow them to function. A receiver determines which distinctions an inquiry retains; preparation can change the regime about which it can ask reliable questions. Those arrangements can themselves become targets of further inquiry. EL pursues this local, constructive connection within specified dynamics, rather than inferring a derivation of fundamental physics from its response experiments. Participation helps specify and produce an experimental situation; it does not give the investigator discretion over whether the resulting claim survives its test.

2.3 Hartle: what makes regularity and its use physically possible?

Gell-Mann and Hartle place observers, apparatus, records and observed systems within one physical account. Their information-gathering and utilising systems exploit regularities, form predictions from records and act within the histories the theory describes. The correlations enabling such activity are not an infallible channel supplied from outside the world. Where Wheeler seeks an origin of physics in parti-

curacy, this account works within quantum mechanics to explain histories in which observers and records can arise. These are distinct explanatory projects. [2,26]

Hartle's account of quasiclassical realms asks how extensive approximate classical predictability arises within quantum mechanics. Suitable coarse-grained histories, including averages of conserved or approximately conserved densities, support effective equations under particular physical conditions. Both dynamics and state enter the explanation. In *Classical Equations for Quantum Systems*, Gell-Mann and Hartle further show that decoherence alone does not suffice for nearly deterministic behaviour: the retained variables need adequate effective inertia in the presence of fluctuations. Coarse graining, noise, dissipation and effective dynamics must be considered together. [3,17]

In *Why Our Universe Is Comprehensible*, Hartle makes the knowing system's own dependence explicit. Exploitable regularities are conditions for such systems to evolve and function. Human observers are phenomena within a quasiclassical realm, not external choosers whose preferences bring that realm into existence. The argument is conditional on the existence of information-gathering systems, not a claim that every possible universe contains them. [24]

This supplies a physical explanatory demand complementary to the justificatory one: **account for the arrangements that make a regularity and dependable access to it possible**. Here two questions must remain distinct. What sustains the relevant physical organisation? And what makes an observation or description adequate for a particular use? The matrix structural investigations address the first; the circuit information challenge and qualification studies address the second; preparation experiments connect them.

The models below already supply classical dynamics and numerical access. They investigate local conditions within that setting, rather than deriving it from quantum cosmology. Reduction can help make effective prediction possible while excessive reduction can discard a distinction needed for a task. These are compatible concerns, not identical mechanisms: the circuit information restriction is not quantum decoherence. Their common scientific demand is to establish which description is adequate under which physical and observational conditions.

2.4 When a condition becomes an object of inquiry

Kant's demand for warrant, Wheeler's account of participation and Hartle's physical explanation are not interchangeable doctrines. Together they direct attention to claims that answer to their objects through arrangements within the world, including arrangements investigators choose and make. Physical production does not establish warrant; warrant does not produce a missing instrument, state or intervention. The constructive question is which of those arrangements can themselves be investigated, maintained or changed.

"Condition" must therefore not conceal differences of kind (Table 2). A support rule can bind the justified use of a package without marking a boundary of physical possibility. An acceptable error is a task commitment, not a property of the dynamics alone. A physical intervention can nevertheless change whether those commitments are met.

Kind of condition	Example in this work	What satisfying it establishes
Physical	A prepared state or persistent dynamical organisation.	The arrangement required by the proposed response has been realised.
Informational	A retained history or an available state description.	Specified distinctions are available to the predictor or decision procedure.
Evidential	A supported domain, calibrated uncertainty and prospective qualification.	The evidence warrants the particular use under declared assumptions.
Task-related	An acceptable response, preservation constraint or effort limit.	The use meets the stated objective and constraints.

Table 2: Conditions with different roles. These distinctions organise tests, not four independent physical variables. In §5.2, a physical preparation changes a measured coordinate so that an existing evidential requirement is satisfied for specified tasks. That does not make the support boundary a boundary of attainable response.

A bounded account can legitimately leave most background conditions fixed. EL makes a selected condition explicit when its adequacy or availability becomes consequential. It does not require every experiment to explain its entire enabling world. The empirical connection is then precise: specify the requirement, test whether it is met, and test whether another operation can supply it without invalidating the intended use.

3 Response laws and scientific inquiry

3.1 A relation, a package and a task

A response law relates a specified intervention to a measured consequence under declared conditions. Let z be a description formed from causally available observations, a a delivered intervention and r_τ the response at horizon τ . A candidate model supplies a prediction

$$\hat{r}_\tau = F(z, a).$$

Its experimental meaning also requires the prepared medium and operating domain D , retained history H , intervention and delivery specification A , receiver R , and horizon τ . Recording these conditions does not assert that all five must be independently varied in every experiment. A command-to-response model may combine delivery, evolution and observation; identifying that composite does not automatically identify its factors.

The term *empirical lawhood* denotes a qualified standing for a response relation: evidence supports its repeatability and prospective use across specified interventions or preparations within an explicit scope. It is more demanding than fitting a realised outcome, and narrower than claiming an unrestricted fundamental law. The relation must support testable consequences beyond the data used to construct it; prospective qualification determines how far that standing extends. An exact universal equation is not required, nor is every useful fitted predictor thereby a law.

A **response-law package** combines the relation with its required observations and interventions, supported domain, uncertainty, horizon, refusal rules and evidential record. A **task** specifies acceptable consequences and constraints. The same package can support several tasks, but each proposed use must meet its requirements. Declining an action means that the procedure does not justify it; the system continues evolving, and unchanged evolution is not safety by definition.

The preparation experiment provides a concrete example. Its 24-coordinate description includes a backward rate of reference force. The downstream relation predicts the measured difference between matched opposite-force branches. The package permits use only inside its fixed support and other qualification

requirements; the tasks additionally constrain response and preservation. Negative preparation moves the force-rate coordinate inside support. The coefficients and evidential rules remain unchanged. This is a physical change satisfying an existing condition of use, not a retrospective widening of the model's domain (§5.2).

3.2 From a failed requirement to another scientific task

An unsuccessful use poses a discriminating question. Was the relevant distinction unobserved, the response inaccurately predicted, uncertainty unqualified, delivery unavailable, or the requested effect absent from the tested action set? Additional samples, another sensor, a different preparation and a changed objective address different obstructions. A richer model is not the universal remedy.

For decision, an action existing for each possible behaviour is different from one action working across the behaviours the observer cannot distinguish. Data informativity formalises important instances of this distinction. [4] The intended action must also satisfy preservation, effort, timing, delivery and uncertainty requirements. Predictive accuracy alone is not their substitute.

A required observation or preparation can become the output sought from another operation. The **recursive step** is to pose that prerequisite as a further problem with its own conditions. This is an organisation of inquiry: its occurrence does not imply successful repair. A **constructed connection** requires evidence that the upstream operation supplies the downstream requirement. **Qualified composition** further requires that the joined use preserves the relevant uncertainty, timing, resource and task commitments. Mere mutual influence, repeated equations or matching labels establish none of these connections.

A fixed preparation recipe may supply a required condition without being a learned law forecasting future applicability. A forecast may supply a useful input while requiring new calibration of the composition. Both occur below. Recursion ends operationally at available capabilities, resource limits or an unresolved obstruction; it does not certify its own premises. The studies developed through exploration, diagnosis and bounded prospective tests, not through one frozen autonomous algorithm.

3.3 Discovery as an application of the framework

Open-ended discovery need not begin with a fixed engineering target. It can begin with an organisation to understand, competing explanations to distinguish, or a description whose adequacy is uncertain. A locally precise question can guide the next experiment without fixing the ultimate destination of the investigation. The object sought may be an explanation or a newly resolved distinction, not an immediately controllable response.

EL structures that search by asking what an apparent success or failure actually establishes. The search can change not only its parameters but its objects: retained history, measured directions, preparation, response horizon or the requirement imposed on a model. A failed structural criterion invites a distinction between disappearance of an organisation and inadequacy of its detector; a failed forecast invites separation of inaccessible information, inaccurate dynamics and unqualified uncertainty. Each alternative calls for a different investigation, not indiscriminately more computation.

The matrix programme provides concrete instances. The isolated geometric event prompted conditional-future tests that distinguished a structural excursion from the committed object initially sought. A separate algebraic analysis examined what survived a scalar detector's failure. The transient analysis separated a predictable intervention-induced increment from the uncertain baseline that obstructed absolute prediction. These were changes of scientific question as well as measurements within a medium (§§4.2–4.4). [19–22]

The programme was a human-directed application and iterative development of EL, not the execution of a fully specified autonomous discovery algorithm. Its substrate-specific findings have two roles: results about their experimental systems, and records of inquiry organised around explicit distinctions and dependencies. A finding need not be substrate-general to count as an output of such a method. Conversely, productive use and co-development do not independently validate a frozen method or establish superiority over alternative

search procedures. The exploratory history and the later prospective tests retain their different evidential roles.

3.4 Relationship to established methods

Predictive-state and input–output approaches organise histories by their future consequences; projection methods explain how reduction introduces memory. [5,10] Data informativity separates identification from some control objectives. [4] Assume-guarantee contracts and causal abstraction formalise compatibility and error across connected descriptions. [6,7] Decision-focused learning and calibrated prediction distinguish fitting error from the requirements of use. [8,9] Cartwright’s nomological machines concern arrangements that sustain regularities. [11]

EL uses these resources when the relevant descriptions and enabling conditions are partly unknown and potentially producible. Its organising contribution is to make an inadequately supplied requirement a discriminable scientific target, and to test whether another operation supplies it. A conventional identifier, model-predictive controller, lookup table or calibrated regression can provide a component. New estimators are neither required at every stage nor excluded. The scientific contributions are the response findings and connections established; the broader methodological hypothesis is that organising inquiry around such requirements can make the acquisition and reuse of capability more effective. A general efficiency advantage remains to be measured.

4 What descriptions preserve—and what they do not

The first studies report scientific findings about information, structure and prediction, before testing whether particular conditions of use can be supplied. They are outputs of the developing inquiry as well as tests of particular propositions within it. An analytical bound, prospective numerical challenges, selected events and retained trajectories retain their distinct evidential scales; they do not form a pooled experimental population.

4.1 Reduced circuit observations and consequential hidden differences

Consider two equal-capacitance circuit preparations with decay times 1 and 4. Initial voltages $(1, 1)V_0$ and $(0.5, 1.5)V_0$ share a mean and total charge, but under unchanged zero-input evolution their subsequent means separate by up to $0.11812V_0$. A deterministic predictor given only the shared initial mean must return the same prediction for both. The elementary inequality

$$\max\{\|\hat{y} - y_1\|, \|\hat{y} - y_2\|\} \geq \frac{1}{2}\|y_1 - y_2\|$$

therefore gives worst-case error at least $0.05906V_0$. No estimator restricted to that identical input can recover the absent distinction. Whether this prevents a decision is separate: both responses may be acceptable, or a different action may work for both states. [18, E1.1]

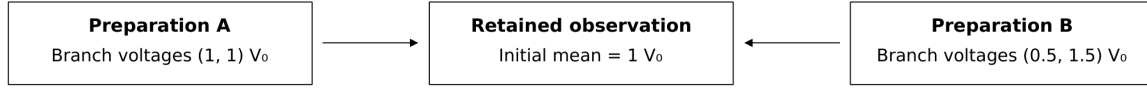
A prospective numerical challenge examined resistor–capacitor ladders across three component-disorder families and 64-, 128- and 256-cell representations. The observer retained eight spatially averaged voltages over specified histories. Hidden-state pairs were nominated before their future responses were checked by a separately derived numerical implementation. The numerical criterion was matching within a declared observation tolerance, not exact identity of predictor inputs. Without a specified quantisation or noise channel, near-equal real-valued observations do not imply impossibility for every unrestricted predictor. The exact bound above is not silently transferred to this numerical setting. [18, E1.2–E1.3]

Across 36 independent seed blocks, 2,319 nominations were confirmed: 795 nonadverse, 625 response-adverse without changing the decision, and 899 adverse for both response and decision (Fig. 1). All declared family/size/endpoint comparisons contained adverse witnesses. These are targeted-search counts, not their prevalence in ordinary operation. Decision-adversity concerns the tested action and criterion, not proof that no satisfactory common action exists. [18, E1.3–E1.4]

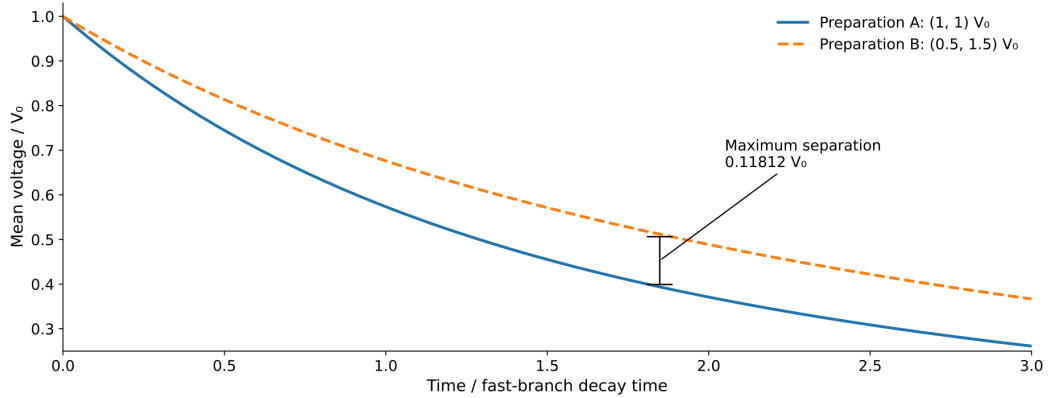
History matters. The entered challenge found 697 response-adverse pairs among 864 nominations at depth zero, and 46 among 188 at depth four. None of the 268 entered pairs at depths eight and nine–thirteen was adverse; depths 16, 31 and 35 were untargetable. This distinguishes longer histories that survived the entered challenge from histories the procedure could not test. The contribution is a task-specific information challenge and its observed dependence on retained history, not a universal failure of coarse graining or of every finite memory. [18, E1.4]

What a description must preserve

Same retained initial observation; different subsequent responses



ANALYTICAL ILLUSTRATION



Initial-mean-only deterministic predictor: worst-case error $\geq 0.05906 V_0$

SEPARATE PROSPECTIVE CHALLENGE — 2,319 TOLERANCE-DEFINED COLLISIONS

795

Nonadverse

625

Response-adverse only

899

Response + decision-adverse

The latter two groups total 1,524 response-adverse pairs; 899 also cross a decision boundary.

36 independent seed blocks · targeted challenge, not prevalence

Figure 1: What a description must preserve. The curves are the exact analytical illustration, not acquired trajectories. The separate numerical challenge contains 1,524 response-adverse pairs, including 899 decision-adverse pairs, nested within 36 independent seed blocks. Each of eighteen family/size/endpoint comparisons contained witnesses in all twelve applicable seed units; the simultaneous one-sided lower bound reported for that recurrence is 0.6123. These comparisons reuse seed units. The history-depth results in the text distinguish adverse, nonadverse and untargetable challenges. [18, E1]

4.2 Geometric organisation without dynamical commitment

The matrix investigations asked whether preparation could change which degrees of freedom furnish a useful response description, predict that change and support subsequent use without refitting. Their medium comprised two coupled triplets of Hermitian 4×4 matrices, X_a, Y_a , evolving under underdamped Langevin dynamics. The finite open extension allows the sectors' couplings to vary independently. Its published six-matrix antecedent supplies the fuzzy-product geometry—a finite matrix representation of geometric organisation—not an EL discovery. [12; 18, E2.1]

The proposed transition would turn internal zero modes into geometric harmonics as a second factor formed. To test it, the feasibility scan sought recurring regimes in which neither, either or both triplets met specified structural criteria. Of 9,126 valid trajectories, 9,125 met neither triplet's criteria and one met only the Y triplet's criteria. The required recurring regimes were not established, so the dependent prediction and nested-control tests did not execute. This is a negative for the tested finite design, not thermodynamic nonexistence of a geometric phase. [19, executive synthesis and §2]

The isolated event nevertheless exhibited an exceptional conjunction of amplitude, approximate angular-momentum closure, spectral separation and persistence. A selected-event study checked replay, same-driver half-step behaviour, amplitude-matched alternatives and simulated futures from seven checkpoints. From each of six checkpoints meeting the original structural criteria, none of 64 futures reached the rolling-window Y -only target before robust return to the rolling-window neither-triplet regime: 0/384 under that stopping rule. Many futures lost and regained instantaneous structural admission; local analysis found several unstable directions. [19, executive synthesis]

The structural conjunction was real under its diagnostics, but it did not establish the dynamically committed object sought. The result favoured an algebraically exceptional, oscillatory excursion over the proposed committed nucleus. Conditional futures from selected checkpoints are not independent discoveries, and the first-passage statistic is not an eventual-return probability. Later long-history samples remain separate from the initial feasibility population. [19; 22, §4.1]

4.3 An approximate algebra survives its scalar detector

A separate retained-history analysis examined whether an effective algebraic structure persisted away from its ideal configuration. The passive operator

$$\Delta_Y(P) = \sum_a [Y_a, [Y_a, P]]$$

identifies directions commuting, or nearly commuting, with the Y sector. At the ideal background $Y_a = r(I_2 \otimes \sigma_a/2)$, its kernel is the expected $M_2(\mathbb{C}) \otimes I_2$: the standard commutant of this construction. [13; 22, §4.2]

At a selected nonideal excursion, the three low nontrivial eigenvalues were approximately 0.1404, 0.1457 and 0.2263, followed by a faster band beginning near 8.34. The fitted slow subspace remained within a maximum principal angle of 0.740° of an $M_2(\mathbb{C})$ factor, with associative leakage about 2.92×10^{-4} . That approximate organisation survived while the scalar canonical-closure diagnostic crossed its 0.30 threshold. Positive slow rates, and their lifting under the joint-sector operator, prevent identifying an exactly conserved subsystem. The finding is persistence of an approximate factor despite failure of a particular detector. [21, §5.1; 22, §§4.2–4.3]

Prediction supplied another distinction. Across 64 histories after removal of the annealing preparation, median history-mean frozen- Y loss at the 32-tick horizon was approximately 2.4×10^{-4} for the full-map aggregate, 0.420 for slow directions and 3.714 for the worst resolved direction. Sixty-three histories had mean worst-direction loss above the no-change reference of one. Aggregate accuracy therefore did not establish uniformly adequate directional prediction. [22, §4.4]

These losses are normalised by actual increments. Weak directions require the specified resolution floor and an assessment of absolute task-scale error before a control consequence follows. The comparison does not itself establish failure of a downstream task. The passive probe equation also used the supplied double-commutator family: successful prediction from that operator is not independent discovery of an endogenous law. The structural and directional findings remain selected-event or retained-history analyses, not prospectively discovered prevalence. [18, E5.7; 21, §5.1]

4.4 Accurate intervention effects and an uncertain absolute future

Endogenous force-response studies made the dynamical question explicit. A full coupled tangent calculation tracked signed finite response more closely than a scalar-curvature reduction in all 6,400 nested comparisons across 32 simulation preparations. Evaluated along the realised future path, it explains measured response but is not a forecast available at the time of choice. The earlier absolute-endpoint development task remained unsupported. [21, §5.2]

A later analysis of retained trajectories found a compact approximation to preparation-induced change. Its fixed radial surrogate was

$$q'' + q' + kq = u(t), \quad k = (88/9)^2.$$

Here q is a surrogate amplitude and u the prescribed preparation waveform. Displacement, velocity and strictly backward differences provided four kernel channels; pre-preparation features predicted their weights in the 24-coordinate description. Frequency and damping were prescribed, not fitted discoveries. Across the tested schedules, the model reduced description-change mean squared error by 93.0% relative to no change, and induced-response-change error through the unchanged response model by 90.1%. It beat a matched non-restoring surrogate on withheld-duration, recovery-schedule and late-time tests. Separate endpoint maps remained more accurate on their represented menu: the radial model's description-change MSE was 41.5% higher there. [20, §1]

The successful prediction concerned an increment. Write the prepared description as

$$z_p = b + \Delta_p, \quad \hat{z}_p = \hat{b} + \hat{\Delta}_p,$$

where b is the simultaneous unchanged-preparation baseline. In the reported response-error decomposition, the squared propagated baseline contribution was about 781 times the transient contribution. Accurate intervention-induced change barely improved the full absolute forecast. Substituting the measured future baseline reduced error, but this was an evaluator-only diagnostic, not available information. The full forecasts still placed all ten observed support failures inside support. [20, §2]

The positive finding is substantial structure in the intervention's effect. The obstruction is failure to locate the absolute state needed for subsequent use. Knowing how an action changes a system can therefore be easier than knowing where that system will be well enough to warrant another action. This retained-data analysis, including held-out comparisons on previously examined preparations, did not provide fresh confirmation or repair the failed prospective preparation claim.

5 Supplying conditions for use

The next studies establish different connections: a forecast supplies a description before preparation, and a physical preparation makes an intact package usable after the prepared state is observed. Neither connection is inferred merely from a good fit; each was prospectively tested under its own qualification procedure.

5.1 Forecast-mediated reuse

An upstream predictor U forecast the 24-coordinate description after an assigned preparation p . An existing response model F then used that description to choose assays:

$$\hat{z} = U(z_0, p), \quad \hat{r}(a) = F(\hat{z}, a).$$

Each episode received unchanged evolution or one of four signed coupling excursions. Preparations were assigned, not adaptively selected to improve applicability. The coefficients of F stayed fixed across preparations; the composition received separate uncertainty calibration. Thus the construction reused a response relation within a newly qualified composition, rather than automatically transporting an unchanged guarantee. [18, E2]

Actions were signed force pulses of magnitude 8 or 16 along either of two directions. Each choice specified a paired assay: opposite-force branches began from the same prepared state with matched future noise, and response was half their output difference. Matched zero-force evolution supplied preservation checks on both signed branches. Two future noise draws and two numerical resolutions evaluated each assay. This is a controlled response comparison, not single-trajectory steering to an absolute endpoint. [18, E2.2]

The description included state and momentum summaries, reference force and curvature information, and backward rates. These quantities were available at the observation time inside the simulator; their equivalent acquisition from ordinary physical sensors was not established. Preparation identity entered U , not F . Point-response regressions were affine at fixed preparation, with preservation predictions clipped separately. [18, E2.3–E2.4]

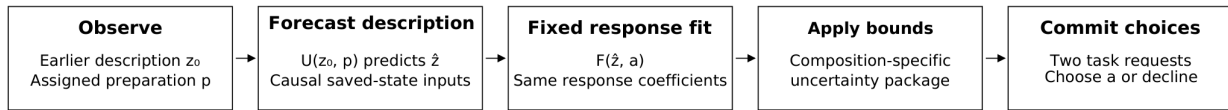
Two task families used the same response model, with longitudinal target ranges, transverse limits and preservation requirements. Predictions and choices were sealed before preparation. Observing the prepared state could cancel a choice, but could not revise its prediction or select a replacement (Fig. 2). Calibration used 32 separate episodes. The 30th ordered complete-episode residual score supplied the declared 90% marginal boundary under the specified exchangeability assumptions; this is not coverage conditional on every state or accepted action. [8,14; 18, E2.5–E2.6]

Both tasks succeeded in **61 of 64 fresh test episodes**, with a one-sided 95% lower bound of 0.8833. No episode contained a false admission; the corresponding one-sided 95% upper bound was 0.04573. All 125 delivered task assays succeeded. Three tasks on three distinct episodes were declined before preparation, and no committed assay was subsequently cancelled. All 64 episodes remain in both denominators. [15; 18, E2.7]

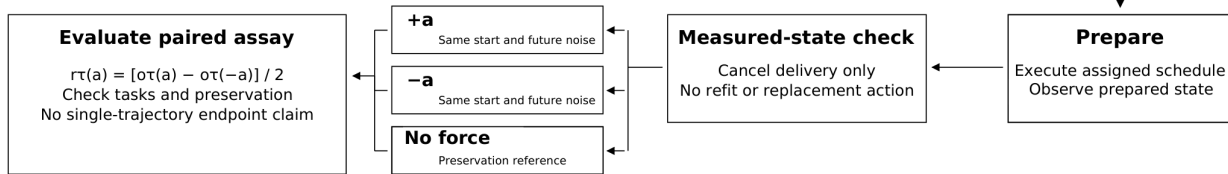
A predicted description supports later use

32 calibration episodes · 64 new prospective episodes · fixed response coefficients

BEFORE PREPARATION: FORECAST AND COMMIT



SEALED CHOICES — PREPARED-STATE OBSERVATIONS CANNOT REVISE THEM



125 successful delivered assays · 3 choices declined before preparation · 0 later cancellations

PROSPECTIVE USE

61 / 64

Both tasks succeed

One-sided 95% lower bound: 0.8833

0 / 64 false-admission episodes

One-sided 95% upper bound: 0.04573

RESPONSE PREDICTION

Mean response MSE / 0.01²

Composed 0.01865547

Cached 0.02574974

Direct 0.01994648

27.55% less loss than cached

Direct comparison: descriptive only

COMPOSED VERSUS CACHED

Joint-task outcomes

Both methods meet both tasks 60

Composed alone meets both 1

Cached alone meets both 0

Neither method meets both 3

Paired one-sided $p = 0.5$

2 / 128 choices changed

Prediction gain is not a demonstrated gain in joint-task success.

All 64 assigned episodes remain in the denominators · §5.1 and Methods and evidence

Figure 2: A forecasted description supports advance choices. Predictions and task choices are committed before preparation. Prepared-state observations can cancel, but cannot revise or replace, those choices. The fixed response coefficients are used through a separately calibrated composition. All 64 assigned episodes remain in the success and false-admission denominators; tasks, future draws and numerical branches do not multiply independent units. Improved prediction is distinct from demonstrated task superiority. [18, E2]

Response MSE was 27.55% below the cached preparation-specific predictor and, descriptively, 6.47% below a direct predictor. Task superiority was not established: cached and composed methods succeeded jointly on 60 and 61 episodes, respectively; the single favourable discordance gave predeclared paired $p = 0.5$, and

only two of 128 choices changed. The achieved result is prospective use through a forecasted input and fixed response coefficients. It neither establishes comparative task advantage nor shows that this factorisation is indispensable. [18, E2.7]

5.2 Preparation-mediated applicability

The second construction held the **entire response package fixed**: coefficients, supported domain, calibration, intervals, action set and refusal rules. It asked whether preparation could supply a state in which that package permitted valid, useful use. Earlier screens had either insufficient improvement headroom or no candidate meeting stricter entry requirements. The successful experiment instead used a separate population deliberately constructed near a fixed support boundary; it did not reclassify those earlier negatives. [18, E3.1–E3.6]

A nominated negative preparation qualified on eight new episodes, excluded from confirmation. Thirty-two further episodes compared equal-duration unchanged evolution H, selected negative preparation N and positive preparation P. Each began from the ideal state, using one previously selected bath waveform plus independent Gaussian perturbations scaled by 0.002 until the prepared-state observation. Their variance was 4×10^{-6} of ordinary variance; subsequent response tests used fresh full-strength noise. The independent perturbations are conditional on one selected recipe, not replication across independently selected recipes or solvers. [18, E3.6–E3.7]

Preparation was selected at tick 4096. The prepared description was observed at tick 4496, after which the unchanged package predicted responses and sealed assay choices before outcomes through tick 4688. Unlike §5.1, these choices used the observed prepared state. Each episode supplied 256 request pairs: 8,192 pairs nested in 32 confirmation episodes. The two tasks could choose different paired assays. Joint success is not simultaneous actuation or shared-resource compatibility on one trajectory. [18, E3.7]

Under the frozen primary-resolution validity rule, N passed on 32/32 confirmation episodes, compared with 0/32 for H and P. Both tasks succeeded within the package's requirements on 7,965/8,192 request pairs under N, versus none under H or P. No false admissions were observed. The paired episode-validity comparison had 32 favourable discordances and none reversed, giving exact one-sided $p = 2.3283 \times 10^{-10}$. Zero false-admission episodes yields a one-sided 95% upper bound of 8.94% under the stated independent, identically distributed episode model—not a per-request safety guarantee. [15; 18, E3.9 and E3.14]

The mechanism identifies what the large result means. A fixed support face bounds the recent rate of reference force along the second response direction, measured by a backward 32-tick difference. Its upper boundary is 252.063 model force per model time. H left this coordinate around 259.8–261.4, N moved it to 250.4–252.0, and P moved it to 275.9–277.5. All three passed the other six checks: numerical response agreement, point accuracy, interval width, preservation, preparation work and realised interval coverage. H and P were declined because they missed the existing domain restriction (Fig. 3). [18, E2.3 and E3.8–E3.10]

An authenticated post hoc test bypassed only support while preserving coefficients, widths, requests and the choice rule. Successful pairs became **7,916 under H, 7,965 under N and 7,952 under P**. The N-minus-H difference was 49 pairs, or 0.598 percentage points, compared with 97.23 points under the intact package. An outcome-visible search found a suitable action for both tasks on every pair under every preparation. These are diagnostic comparisons, not qualified replacement policies or permission for unsupported use. They establish that the dominant effect was eligibility for unchanged knowledge, not creation of previously unavailable task capacity. [18, E3.14]

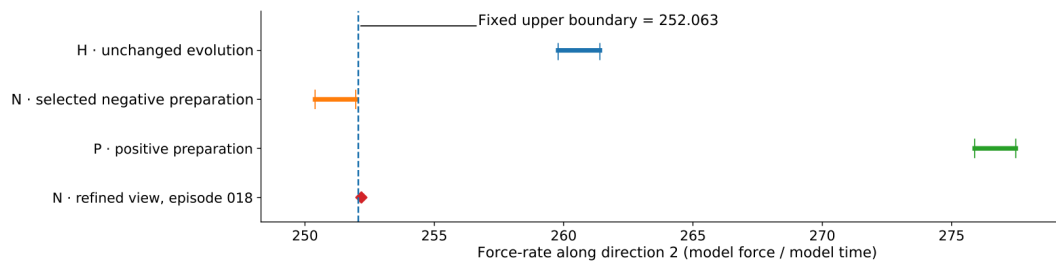
The numerical margin was narrow. N lay inside support in both resolutions on 31/32 episodes; a conservative discrepancy-based margin retained 27/32. Counting exclusions as zero retained 7,719 and 6,720 successful pairs, respectively. The conservative result is 82.03% of all requests, compared with zero under unchanged evolution. These post hoc checks retain a substantial result while exposing sensitivity near the boundary; they do not establish independent-solver reproduction. [18, E3.14]

Preparing conditions for an unchanged response package

Preparation changes the state. The entire response package stays fixed.

32 confirmation episodes · 8,192 request pairs · 8 qualification episodes excluded

A FIXED DOMAIN BOUNDARY AND PREPARED-STATE RANGES



Ranges are reported minima/maxima, not uncertainty intervals. Diamond: one refined-view value.

B BENCHMARK SUCCESS VERSUS AVAILABLE OPPORTUNITY

Count / 8,192 request pairs

Frozen supported policy — PRIMARY

	H	N	P
Frozen supported policy — PRIMARY	0	7,965	0
Support alone bypassed — POST HOC	7,916	7,965	7,952
Outcome-visible oracle — POST HOC	8,192	8,192	8,192

Bypass N — H: 49 pairs (0.598 points). Primary gain: 97.23 points.

Bypass is diagnostic only; the outcome-visible search is not a deployable selector.

C SUCCESSFUL USE UNDER STRICTER NUMERICAL CHECKS

	N episodes / 32	Covered pairs / 8,192
Frozen primary view	32/32	7,965 (97.23%)
Both-view support · post hoc	31/32	7,719 (94.23%)
Conservative envelope · post hoc	27/32	6,720 (82.03%)

Excluded episodes count as zero successful use; every assigned episode remains in the denominator.

Paired response assays · separate branches for the two tasks · §5.2

Figure 3: Preparation changes applicability without changing the package. Panel A shows primary-resolution minima and maxima, not uncertainty intervals; the refined-resolution exception is separate. Panel B distinguishes successful qualified use from post hoc support-bypass and outcome-visible opportunities. Panel C reports successful pairs satisfying the relevant package checks, including realised coverage, under increasingly conservative numerical rules. Excluded episodes count as zero and all 32 confirmation episodes remain in the denominators. Eight qualification episodes are excluded. Each request pair comprises separate paired assays, not simultaneous actuation. [18, E3.14]

This is a qualified fixed recipe that repeatedly supplied a condition for using an unchanged package in a local population. It is not a newly calibrated forecast of preparation-dependent applicability. Preparation was not shown necessary, cheapest or preferable to independently qualifying broader support. Nor was this construction joined to §5.1 in one qualified controller. Its value is the tested connection itself: action made the requirements of existing response knowledge attainable without altering that knowledge's commitments.

6 Different systems, different requirements

The wider studies investigate whether the same distinctions matter in other simulations and physical records. They do not supply a pooled success rate, a universal response equation or evidence that an individual law transfers across substrates. Each contributes a particular finding or tests a particular connection (Table 3). [18, E4–E5; 22, §3]

System and evidence	Result	Role in the argument
Physical cortical records; 268 cells	Logged safe options in 162 cells did not establish successful selection.	Available opportunities and a justified selector are different requirements.
Optical mirror; three held-out blocks	Correct/reversed-action RMSE: 1.325/5.368 Hz; the underlying amplitude effect was previously published.	Action sign has a measurable consequence; using that effect is not discovering it anew.
Physical battery records; 44 cells	Improved mean prediction; nominal 90% cell-level voltage/capacity error envelopes covered 43.2%/93.2%.	Accurate means do not establish adequate uncertainty for the intended measurement.
MAST-U divertor archive; two discharges	Doubling modulation frequency reduced requested-flow/front gain to 0.589/0.594 of its earlier value.	Commanded drive is not measured delivery; delivered flow and complete sink accounting remain unresolved.
Gym-TORAX and PyBaMM; 18 and 14 fresh efficacy preparations	Selected-action tests met their respective effect requirements.	Local prospective simulated usefulness is possible without demonstrated physical transfer.
Reduced PyBaMM description; twelve fresh preparations	Four-coordinate prediction passed 479/480 supported continuation checks, with one thermal failure.	A useful reduction is task- and support-dependent; this description did not itself enter prospective control.
Reactor simulator; 96 qualification and 64 prospective episodes	Forty-seven atlas coordinates qualified; five context-horizon pairs supported use.	Qualification can support useful service while sacrificing service frequency relative to comparators.
MAST heating-TORAX bridge	Physical classifier: 100/170, below criterion; corrected generic replay declined intervention.	Physical association and simulator ordering did not establish a useful joined transfer construction.

Table 3: What the cross-system evidence contributes. Cells, blocks, discharges and simulation episodes retain their original acquisition hierarchies. Multiple requests or numerical views do not enlarge those populations. Physical-record analyses are not new prospective physical controllers. Results and qualifications are detailed in the methods companion. [18, E4–E5; 22, §§3.2–3.4]

The simulated selected-action studies show bounded local use. Gym-TORAX retained useful effects across declared numerical views despite unresolved magnitude differences; the PyBaMM benefit was closely tied to integrated current. The separate reduced PyBaMM description retained one thermal failure. More elaborate descriptions incurred support exclusions and did not simply dominate: this concerns finite-data estimation and qualification, not a claim that additional measured information is intrinsically harmful. [18, E5.5–E5.6; 22, §3.3]

In the reactor simulator, a finite interval response atlas qualified 47 of 72 context/action/horizon coordinates. Five of twelve context-horizon pairs entered prospective use, supplying 636 successful deliveries among 640 planned uses across 64 scenario episodes; four interventions were declined by the thermal guard. The seven other pairs did not become failed use trials. A separate same-episode counterfactual audit found fewer safely served requests for the conservative policy than for every comparator, but no observed false admissions for that policy. The result supports useful local qualification, not the rejected service-gain claim or an optimal service-risk frontier. The paired effects were resolved at millikelvin scale inside the simulator; equivalent online physical resolution was not demonstrated. [18, E4]

These diagnoses change the next scientific question. Missing measurement resolution calls for a different means of observation; failed uncertainty requires new qualification rather than denial of the mean effect; unavailable delivery cannot be repaired solely by a better response fit. The records document human-directed application of these distinctions across systems, not a measured general reduction in the cost of scientific search.

7 From dependencies to cumulative capability

7.1 What has been established

The findings concern three distinct achievements. First, particular scientific structures and obstructions were identified: tolerance-defined circuit witnesses, geometric organisation without the expected commitment, approximate algebra beyond a detector's threshold, unequal directional accuracy, and the separation of intervention effects from an uncertain absolute future. Each stands on its own evidence. They are not interchangeable confirmations of a single universal proposition.

Second, selected prerequisites were supplied. A forecast supported advance choices through fixed response coefficients with new calibration of the composition. A physical preparation made an entirely unchanged package usable in a constructed local population. The difference matters: one supplies an informational input before preparation; the other supplies a physical condition whose adequacy is then observed. Together they show that some requirements of usable knowledge can be investigated and produced. They do not establish that every requirement can be supplied or that these two constructions already form an integrated chain.

Third, the programme documents the human-directed use and development of an organisation of inquiry around explicit dependencies. The matrix findings contribute both as results about their substrate and as outputs of an identifiable course of investigation. Their production does not by itself establish the comparative efficiency of the framework. A failed study contributes when it discriminates among explanations or rules out a proposed connection. Failure alone does not corroborate the programme.

7.2 Dependence, production and objectivity

An analytical constraint, an empirical outcome and an evidential rule have different force. Under the circuit illustration's identical-input restriction, the error bound follows necessarily. The preparation study instead demonstrates satisfaction of a particular package requirement. Its support boundary is binding for that package's qualified use, not for every physically possible route to the desired response. Neither result implies that a system can investigate or repair every condition on which it depends.

The philosophical proposal is that the conditions of knowing and acting belong among their objects of inquiry. An instrument, record or preparation is not merely an inconvenience on the way to a description from nowhere. It is part of the organised relation through which a claim can be tested and an action can have its intended consequence. Some such relations can become the means of establishing others. The empirical work gives this proposal substance by distinguishing conditions, demonstrating selected connections and locating failures; it does not derive the proposal from a pooled count of successful episodes.

This also makes the local relation between participation and objectivity precise. Wheeler's question returns at the level of chosen apparatus and registered consequences. [25,26] Preparing a state changes what can be investigated or done, but does not make a prediction true by declaration. The unchanged prediction can still fail. Objectivity requires that the arrangement and its contribution be accountable to consequences, not that the investigator pretend to have made no difference. Intended agency is one kind of causal participation; unintended disturbance, consumption and decay are others. A complete theory of subjectivity is not required to test these dependencies.

7.3 A civilisational response-law engine

A **civilisational response-law engine** would organise scientific contributions by the consequences they support, the conditions they require and the evidence connecting them. A missing observation or preparation becomes a target another contribution might supply. Compatibility is tested at the connection, including uncertainty, timing, costs and resource use, rather than inferred from matching names. Contracts and compositional abstraction provide formal resources for parts of this task. [6,7]

A materials investigation, for example, could establish a preparation that places a microstructure inside a device model's supported domain. The device could supply a measurement required to investigate another

material. Each connection would carry fabrication variability, measurement resolution and qualification obligations. This is an illustrative application, not a completed experimental chain. What would accumulate is not only descriptions of responses, but tested means of making additional questions answerable. The next question could be chosen for its scientific interest, for an intended application, or because its answer would enable another investigation.

Three forms of accumulation remain distinct: applying a procedure to another system, reusing an established response description for another request, and reducing the new evidence needed for a further capability. The present work includes instances of the first two. Establishing the third requires measuring acquisition and qualification costs, including failed searches and the cost of making components compatible. Preparing a system to preserve an old package must compete with qualifying a broader package; preserving a preferred model at any cost is not the objective.

The engine can fail concretely. It may explain failures only retrospectively, require inaccessible measurements, spend more on qualification than reuse saves, or join components in ways that invalidate their guarantees. Its development is a collective scientific undertaking. Laboratories, communities and disciplines can contribute different components, and simpler qualified methods should be able to replace elaborate ones without losing the meaning of the evidence. Neither a single autonomous agent nor a universal controller defines the architecture.

7.4 Scientific agency and the direction of inquiry

A further aim is to expand and preserve **scientific agency**: the capacity of people and communities to formulate questions, choose investigative paths and redirect them in light of evidence. Access to an answer is not the same as the ability to originate and pursue an inquiry. Making alternatives, dependencies and limits intelligible is therefore a proposed function of the engine, not merely an explanation appended to its outputs.

The intended relationship resembles a cockpit rather than a final approval button. Automated components may propose hypotheses, run calculations and select experiments within an agreed remit. People need not inspect every operation, but consequential choices must remain understandable and revisable at the relevant timescale. Changing an informed judgement should be able to change the inquiry's course. Human-in-the-loop participation in this sense is neither protected labour nor a ceremonial veto. A human-directed engine can be the intended design, not an unfinished autonomous one.

Engelbart's joint treatment of people, tools, language, methods and training offers a direct antecedent. [27] Meaningful-human-control research likewise distinguishes responsiveness to human reasons and accountable understanding from mere intervention. [28] In EL, directing inquiry depends on information and means of action whose adequacy should themselves be explicit. This proposed connection has not been tested as human-interface effectiveness or general agency preservation.

The aspiration is principled scientific discovery open to many, not an agenda reserved for those controlling the infrastructure. Automation and concentrated authority are different dimensions; delegated autonomy is compatible with this aim. Accessibility and beneficial civilisational outcomes remain objectives, not demonstrated results. Human choice does not guarantee benefit or confer legitimate authority. Enlarging one actor's control can diminish another's agency; responsible use therefore requires collective judgement about purposes and acceptable harms as well as qualified response knowledge.

The Neptune ambition asks what cumulative dependability would make possible. The inquiry reaches further than that application: the means of knowing and changing a world are themselves among the arrangements that can be known and changed. The empirical contribution establishes some of those dependencies and operations that supply them. The larger scientific prospect is **knowledge that helps produce the conditions of further knowledge, and capability that makes previously inaccessible questions answerable**. The intended engine would enlarge not only what civilisation can accomplish, but who can participate in choosing what it should try to understand and do.

Methods, evidence and availability

Designs, units and timing

The evidence comprises an analytical construction, prospective numerical challenges, selected-event and retained-history matrix analyses, separately qualified prospective interventions, and analyses of physical records. The technical companion [18, E1–E5] provides circuit, forecasted-description, preparation, reactor and wider-system methods. Structural-event, transient and directional-response analyses are documented in [19–22]. The accompanying evidence and provenance note [23] maps these reports to the underlying record identities and preserves differences between exploratory, confirmatory and post hoc analyses.

The circuit bound uses identical predictor inputs; numerical witnesses use a declared tolerance and a separately derived checking implementation of the same physical model. The 2,319 nominations are nested in 36 seed blocks. Conditional futures from selected matrix checkpoints do not enlarge the population of discovered events. Retained-history fits and realised-future tangents do not become prospective forecasts by predicting their recorded outcomes well.

The forecasted-description test retains all 64 assigned episodes in success and false-admission denominators, including pre-preparation declines. Its 32 calibration episodes are separate. The preparation test retains 32 confirmation episodes, excludes eight qualification episodes, and nests all 8,192 request pairs within confirmation. Paired branches, future noise draws and numerical views do not create new independent episodes. Its local population is conditional on the selected bath recipe and perturbation regime.

Qualification and interpretation

For forecast-mediated reuse, the calibration score aggregates required outputs, action pairs, future draws and numerical views at the complete-episode level. The 30th ordered score among 32 calibration episodes implements the stated 90% marginal boundary under the specified exchangeability assumptions. It is not a statewise or accepted-action conditional guarantee. [8,14; 18, E2.6] Point predictions, calibrated widths, deterministic numerical allowances and final decision intervals are distinct quantities; the executed decision records are the appropriate objects for replay. Historical worked-example and interval-reporting issues are documented in the evidence note rather than repaired by inferred operands. [18, E2.8; 23]

For preparation, primary validity, support in both numerical views and the conservative discrepancy margin remain different rules. Excluded episodes contribute zero successful pairs and remain in denominators. Support-bypass and outcome-visible comparisons are post hoc diagnostics, not authorised selectors. The count-based estimates and explicitly labelled exact one-sided bounds are retained; historical bootstrap ranges without a specified confidence level are not presented as 95% intervals. [15; 18, E3.14; 23]

Here *qualified successful use* means that the request is admitted under the specified package checks and its evaluated assay meets the stated outcome requirements. A prediction interval's coverage is one check, not a synonym for task success. *No observed false admissions* is an empirical count, not a universal safety guarantee. The retrospective consistency audit reused the implementing assistant and measurement code; it was not external review or independent reproduction of the physical dynamics. Fixed preparation-nomination scores are not calibrated forecasts of applicability. [18, E3.14]

Human direction and exploratory development

The author directed the evolving research questions and interpreted results with AI assistance in implementation, analysis and writing. The metatheory and experiments informed each other; this history is not an independent test of a method fixed before the entire programme. Within that development, individual prospective studies froze their own predictions, choices and evaluation rules. Later reinterpretation or redirection does not alter those commitments or promote selected-event and retained-data analyses to fresh confirmation. The evidence note separates this account of the research process from the measured outcomes. [23]

Data, code and authorship

All referenced evidence and data are to accompany this preprint in the project repository, github.com/jokebroker/empirical_lawhood. The release is to include the cited experimental records, native trajectories and arrays, fitted models, preparation and observation specifications, decision and evaluation records, analysis code, and the mapping between reported results and their source identities. For third-party records, provenance and access terms remain explicit. The manuscript source bundle contains the text, figures, build materials and documentary sources; it is not a substitute for the accompanying research-data repository.

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References

- [1] Kant, I. (1781/1787). *Critique of Pure Reason*, B19, A84/B116, A158/B197 and B165. English translation by J. M. D. Meiklejohn.
- [2] Gell-Mann, M. and Hartle, J. B. (1990). Quantum mechanics in the light of quantum cosmology. In W. H. Zurek (ed.), *Complexity, Entropy, and the Physics of Information*, SFI Studies in the Sciences of Complexity VIII. Addison-Wesley. Later open deposit: arXiv:1803.04605 (2018).
- [3] Hartle, J. B. (2011). The quasiclassical realms of this quantum universe. *Foundations of Physics* **41**, 982–1006. doi:10.1007/s10701-010-9460-0. Open manuscript: arXiv:0806.3776.
- [4] van Waarde, H. J., Eising, J., Trentelman, H. L. and Camlibel, M. K. (2020). Data informativity: A new perspective on data-driven analysis and control. *IEEE Transactions on Automatic Control* **65**, 4753–4768. doi:10.1109/TAC.2020.2966717.
- [5] Barnett, N. and Crutchfield, J. P. (2015). Computational mechanics of input–output processes: Structured transformations and the epsilon-transducer. *Journal of Statistical Physics* **161**, 404–451. doi:10.1007/s10955-015-1327-5.
- [6] Shali, B. M., Heidema, H. M., van der Schaft, A. J. and Besselink, B. (2022). Series composition of simulation-based assume-guarantee contracts for linear dynamical systems. arXiv:2209.01844.
- [7] Rischel, E. F. and Weichwald, S. (2021). Compositional abstraction error and a category of causal models. *Proceedings of Machine Learning Research* **161**, 1013–1023.
- [8] Angelopoulos, A. N. and Bates, S. (2023). Conformal prediction: A gentle introduction. *Foundations and Trends in Machine Learning* **16**(4), 494–591. doi:10.1561/2200000101.
- [9] Donti, P. L., Amos, B. and Kolter, J. Z. (2017). Task-based end-to-end model learning in stochastic optimization. *Advances in Neural Information Processing Systems* **30**. Open manuscript: arXiv:1703.04529.
- [10] Chorin, A. J., Hald, O. H. and Kupferman, R. (2000). Optimal prediction and the Mori–Zwanzig representation of irreversible processes. *Proceedings of the National Academy of Sciences* **97**, 2968–2973. doi:10.1073/pnas.97.7.2968.
- [11] Cartwright, N. (1999). Nomological machines and the laws they produce. In *The Dappled World: A Study of the Boundaries of Science*, chapter 3, pp. 49–74. Cambridge University Press. doi:10.1017/CBO9781139167093.004.
- [12] Badis Ydri, Ramda Khaled and Rouag Ahlam (2016). Geometry in transition in four dimensions: A model of emergent geometry in the early universe and dark energy. *Physical Review D* **94**, 085020. doi:10.1103/PhysRevD.94.085020. Open manuscript: arXiv:1607.06761.
- [13] Holbrook, J. A., Kribs, D. W., Laflamme, R. and Poulin, D. (2005). Noiseless subsystems for collective rotation channels in quantum information theory. *Integral Equations and Operator Theory* **51**, 215–234. Open manuscript: arXiv:math/0402105.
- [14] Barber, R. F., Candès, E. J., Ramdas, A. and Tibshirani, R. J. (2021). The limits of distribution-free conditional predictive inference. *Information and Inference* **10**(2), 455–482.
- [15] Clopper, C. J. and Pearson, E. S. (1934). The use of confidence or fiducial limits illustrated in the case of the binomial. *Biometrika* **26**(4), 404–413. doi:10.1093/biomet/26.4.404.
- [16] Heidegger, M. (1927). *Being and Time*, §§15–16. English translation by J. Macquarrie and E. Robinson (1962). Blackwell.
- [17] Gell-Mann, M. and Hartle, J. B. (1993). Classical equations for quantum systems. *Physical Review D* **47**, 3345–3382. doi:10.1103/PhysRevD.47.3345. Open manuscript: arXiv:gr-qc/9210010.
- [18] Seneque, G. (2026). *Empirical Lawhood: Methods and Evidence*. Technical companion, 1 October. E1–E5 document the electrical, forecasted-description, preparation, reactor and wider-system studies.

- [19] Seneque, G. (2026). *Geometry Without Commitment*. Structural-event and conditional-future dossier, 29 August. Executive synthesis and §2.
- [20] Seneque, G. (2026). *Baseline-anchored, pulse-timed recursive closure*. 20 September, §§1–2. These sections report completed retained-data transient analysis and baseline diagnosis; later proposed experiments are not results.
- [21] Seneque, G. (2026). *Empirical Lawhood*. Archived matrix-response account, §5: approximate algebraic structure, passive response and realised-future coupled-tangent comparison.
- [22] Seneque, G. (2026). *Empirical Lawhood and Epistemic Control: Scientific synthesis, philosophical argument, and research direction*. 6 September, §§3–4. Supporting matrix and cross-system analyses.
- [23] Seneque, G. (2026). *Empirical Lawhood: Evidence and provenance*. Accompanying note, 4 October. Study-to-source mapping, release assumptions, retained inferential boundaries and historical verification issues.
- [24] Hartle, J. B. (2016). *Why Our Universe Is Comprehensible*. arXiv:1612.01952.
- [25] Wheeler, J. A. (1983). Law without law. In J. A. Wheeler and W. H. Zurek (eds.), *Quantum Theory and Measurement*, pp. 182–213. Princeton University Press.
- [26] Wheeler, J. A. (1989). Information, physics, quantum: The search for links. In *Proceedings of the 3rd International Symposium on Foundations of Quantum Mechanics in the Light of New Technology*, pp. 354–368. A related version appears in W. H. Zurek (ed.), *Complexity, Entropy, and the Physics of Information* (1990), Addison-Wesley.
- [27] Engelbart, D. C. (1962). *Augmenting Human Intellect: A Conceptual Framework*. SRI Summary Report AFOSR-3223, October. Stanford Research Institute. §§I.A, II.A and IV.
- [28] Santoni de Sio, F. and van den Hoven, J. (2018). Meaningful human control over autonomous systems: A philosophical account. *Frontiers in Robotics and AI* 5, 15. doi:10.3389/frobt.2018.00015.